

Modern Algebra

Assignment 12

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Problem 2

We observe that

- $0 * 6 = 0$;
- $2 * 6 = 2$;
- $4 * 6 = 4$;
- $6 * 6 = 6$;
- $8 * 6 = 8$.

Thus, 6 is the unity of this ring under addition and multiplication modulo 10.

Problem 3

Consider the ring of the real valued functions, we denote it by R , and let S be the set of odd functions. Clearly, we can see that S is closed under subtraction, that is, pick two functions $f_1, f_2 \in S$ and see that $f_1 - f_2 \in S$. Next, we show S does not form a ring. Observe that $f_1 \cdot f_2 \notin S$ because it is an even function, so it is not closed under multiplication. Hence, S is not a ring.

Problem 5

Theorem: *If a ring has a unity, it is unique. If a ring element has a multiplicative inverse, it is unique.*

Proof. For the first part, suppose we have a ring R with a unity. That is, we have an element $e \in R$ such that $e \cdot r = r$, where $r \in R$. To prove that the unity is unique assume that we have another element $s \in R$ such that $s \cdot r = r$. Observe that

$$s \cdot e = e$$

$$s \cdot e = s$$

Thus, $s = e$ and the unity is unique.

For the second part, suppose that we have an element $r^{-1} \in R$ such that $r \cdot r^{-1} = r^{-1} \cdot r = e$, where e is the unity. Suppose we have another element t such that $r \cdot t = t \cdot r = e$. See that

$$t \cdot r = e$$

$$t \cdot r \cdot r^{-1} = e \cdot r^{-1}$$

$$t \cdot (r \cdot r^{-1}) = r^{-1}$$

$$t \cdot e = r^{-1}$$

$$t = r^{-1}.$$

Hence, $t = r^{-1}$ and the multiplicative inverse is unique. ■

Problem 6

We observe when $n = 6$

- See that $3^2 = 3$.
- See that $2 \cdot 3 = 0$, $2, 3 \neq 0$.
- See that $2 \cdot 3 = 4 \cdot 3 = 0$, $3 \neq 0, 2 \neq 4$.

No, n is not prime.

Problem 12

Proof. Suppose we have a commutative ring R . Let a, b, c be elements in R , where a is a unit. That is, there is an element $a^{-1} \in R$ such that $a \cdot a^{-1} = 1_R$.

For the forward direction, suppose that $b \mid c$, then there is an element $t \in R$ such that $c = b \cdot t$. See that

$$\begin{aligned} c &= b \cdot t \\ c &= b \cdot a \cdot a^{-1} \cdot t \\ c &= (b \cdot a) \cdot (a^{-1} \cdot t) \end{aligned}$$

Since $(a^{-1} \cdot t) \in R$, c is divisible by $a \cdot b$.

For the backward direction, suppose that $a \cdot b \mid c$, then there exists an element $s \in R$ such that $(a \cdot b) \cdot s = c$. Observe

$$\begin{aligned} (a \cdot b) \cdot s &= c \\ a \cdot b \cdot s &= c \\ a \cdot s \cdot b &= c \\ (a \cdot s) \cdot b &= c \end{aligned}$$

Since $(a \cdot s) \in R$, then $b \mid c$. Therefore, that satisfies the proof. ■

Problem 39

Proof. Suppose R is a ring with unity 1 and a is an element in R such that $a^2 = 1$. Consider $S = \{a \cdot r \cdot a \mid r \in R\}$. We need to show that S is a subring of R . We begin by showing that S is not empty, observe that $a(1)a = a^2 = 1$. Thus, S is not empty. Next, we need to show that it is closed under subtraction, pick $r_2 \in R$ and observe that

$$ara - ar_2a = a(ra - r_2a) = a(r - r_2)a$$

We know that $r - r_2 \in R$, so $a(r - r_2)a \in S$ and S is closed under subtraction. Next, we show it is closed under multiplication. Observe that

$$ara \cdot ar_2a = ar(aa)r_2a = ar(a^2)r_2a = arr_2a.$$

Since $rr_2 \in R$, $arr_2a \in S$ and S is closed under multiplication. Hence, S is a subring in R . ■

For the second question, we showed that S is not empty as it contains 1.

Problem 40

We disprove this statement by showing that R is not closed under multiplication.

Consider two matrices $A, B \in R$. Let $a = 1, b = 0$ for matrix A , and $a = 0, b = 1$ for matrix B .

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Observe the product $A \cdot B$:

$$\begin{aligned} A \cdot B = C &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (1)(0) + (1)(1) & (1)(1) + (1)(1) \\ (1)(0) + (0)(1) & (1)(1) + (0)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

For a matrix to be in R , it must be of the form $\begin{bmatrix} a & a+b \\ a+b & b \end{bmatrix}$, which implies the matrix must be symmetric. In our product, C_{12} is 2 and C_{21} is 0. Since $2 \neq 0$, the product $A \cdot B \notin R$. Thus, R is not a subring of $M_2(\mathbb{Z})$.

Problem 41

We prove that $R = \left\{ \begin{bmatrix} a & a-b \\ a-b & b \end{bmatrix} \mid a, b \in \mathbb{Z} \right\}$ is a subring of $M_2(\mathbb{Z})$.

Proof. First, we show that R is non-empty. Let $a = 0$ and $b = 0$, then $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in R$.

Next, we check for closure under subtraction. Let $A, B \in R$ such that

$$A = \begin{bmatrix} a_1 & a_1 - b_1 \\ a_1 - b_1 & b_1 \end{bmatrix}, \quad B = \begin{bmatrix} a_2 & a_2 - b_2 \\ a_2 - b_2 & b_2 \end{bmatrix}$$

Observe that

$$\begin{aligned} A - B &= \begin{bmatrix} a_1 - a_2 & (a_1 - b_1) - (a_2 - b_2) \\ (a_1 - b_1) - (a_2 - b_2) & b_1 - b_2 \end{bmatrix} \\ &= \begin{bmatrix} (a_1 - a_2) & (a_1 - a_2) - (b_1 - b_2) \\ (a_1 - a_2) - (b_1 - b_2) & (b_1 - b_2) \end{bmatrix} \end{aligned}$$

Since integers are closed under subtraction, let $a' = a_1 - a_2$ and $b' = b_1 - b_2$. The resulting matrix is in R .

Finally, we check for closure under multiplication. Let $A = \begin{bmatrix} a & a-b \\ a-b & b \end{bmatrix}$ and $X = \begin{bmatrix} x & x-y \\ x-y & y \end{bmatrix}$.
The product $P = A \cdot X$ is:

$$P = \begin{bmatrix} a & a-b \\ a-b & b \end{bmatrix} \begin{bmatrix} x & x-y \\ x-y & y \end{bmatrix}$$

We calculate the entries:

- $P_{11} = ax + (a-b)(x-y) = ax + ax - ay - bx + by = 2ax - ay - bx + by$
- $P_{12} = a(x-y) + (a-b)y = ax - ay + ay - by = ax - by$
- $P_{21} = (a-b)x + b(x-y) = ax - bx + bx - by = ax - by$
- $P_{22} = (a-b)(x-y) + by = ax - ay - bx + by + by = ax - ay - bx + 2by$

See that $P_{12} = P_{21}$. Furthermore, observe that

$$P_{11} - P_{22} = (2ax - ay - bx + by) - (ax - ay - bx + 2by) = ax - by$$

This matches P_{12} . Thus, if we let $\alpha = P_{11}$ and $\beta = P_{22}$, then $P = \begin{bmatrix} \alpha & \alpha - \beta \\ \alpha - \beta & \beta \end{bmatrix}$. Since $\mathbb{Z}$ is closed under addition and multiplication, $\alpha, \beta \in \mathbb{Z}$ and $P \in R$.

Hence, R is a subring of $M_2(\mathbb{Z})$. ■